

Angles between lines and planes

Basic skills

Q1. $\underline{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}$, $\underline{c} = \begin{pmatrix} -1 \\ 3 \\ -5 \end{pmatrix}$

a) $\underline{a} \cdot \underline{b} = 1 \times 0 + 2 \times 5 + 3 \times 2 = 16$

b) $\underline{a} \cdot \underline{c} = 1 \times -1 + 2 \times 3 + 3 \times -5 = -10$

c) $\underline{c} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & 3 & -5 \\ 0 & 5 & 2 \end{vmatrix} = \underline{i} \begin{vmatrix} 3 & -5 \\ 5 & 2 \end{vmatrix} - \underline{j} \begin{vmatrix} -1 & -5 \\ 0 & 2 \end{vmatrix} + \underline{k} \begin{vmatrix} -1 & 3 \\ 0 & 5 \end{vmatrix}$
 $= 31\underline{i} + 2\underline{j} - 5\underline{k}$

d) $|\underline{c}| = \sqrt{1^2 + 3^2 + 5^2} = \sqrt{35}$

e) For angle between $\underline{a}$, $\underline{b}$

$$\cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|} = \frac{16}{\sqrt{1^2 + 2^2 + 3^2} \sqrt{5^2 + 2^2}} = 0.794$$

$$\theta = \cos^{-1} 0.794 = 37.4^\circ$$

f) $\underline{n} = \underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 2 & 3 \\ 0 & 5 & 2 \end{vmatrix} = \underline{i} \begin{vmatrix} 2 & 3 \\ 5 & 2 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 3 \\ 0 & 2 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & 2 \\ 0 & 5 \end{vmatrix}$
 $= -11\underline{i} - 2\underline{j} + 5\underline{k}$

Q2. $\underline{r} = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} p_1 + \lambda \\ p_2 + 2\lambda \\ p_3 + 3\lambda \end{pmatrix} = \begin{pmatrix} \lambda - 2 \\ 2\lambda \\ 5 + 3\lambda \end{pmatrix}$

so $p_1 = -2, p_2 = 0, p_3 = 5 \Rightarrow \underline{p} = \begin{pmatrix} -2 \\ 0 \\ 5 \end{pmatrix}$

Q3. $\cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|} = \frac{4 \times 1 + 3 \times 3 + 0 \times -5}{\sqrt{4^2 + 3^2 + 0^2} \sqrt{1^2 + 3^2 + 5^2}} = \frac{13}{5\sqrt{35}}$

$\theta = 63.9^\circ$

Q4. use $\underline{d}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \underline{d}_2 = \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$

$\cos \theta = \frac{\underline{d}_1 \cdot \underline{d}_2}{|\underline{d}_1| |\underline{d}_2|} = \frac{2 - 8 + 3}{\sqrt{1^2 + 2^2 + 3^2} \sqrt{2^2 + 4^2 + 1^2}} = \frac{-3}{7\sqrt{6}}$

$\theta = 100^\circ$

$\theta = 80^\circ \leftarrow \text{acute angle between planes}$



Competency skills

Q5. $\underline{d} = \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}$ $\underline{n} = (5, 7, 10)$

a) $\underline{d} \cdot \underline{n} = 2 \times 5 + 4 \times 7 + 1 \times 10 = 48$

$$|\underline{d}| = \sqrt{2^2 + 4^2 + 1^2} = \sqrt{21}$$

$$|\underline{n}| = \sqrt{5^2 + 7^2 + 10^2} = \sqrt{174}$$

b) so β is between vectors

$$\cos \beta = \frac{\underline{d} \cdot \underline{n}}{|\underline{d}| |\underline{n}|} = \frac{48}{\sqrt{21} \sqrt{174}}$$

$$\text{so } \beta = 37.4^\circ$$

$$\text{so } \alpha = 90 - \beta = 52.6^\circ$$

for between plane and line

Q6. $\underline{r} = \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$ Plane (5, 4, 2),
Points: (-2, 6, 10),
(0, 1, 4)

so in-Planar: $\begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \\ -2 \end{pmatrix}$
 $\begin{pmatrix} -2 \\ 6 \\ 10 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \\ 6 \end{pmatrix}$

So normal is

$$\underline{n} = \begin{pmatrix} \underline{i} \\ \underline{j} \\ \underline{k} \end{pmatrix} \times \begin{pmatrix} \underline{s} \\ \underline{s} \\ \underline{6} \end{pmatrix} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \underline{s} & \underline{s} & \underline{-2} \\ \underline{-2} & \underline{s} & \underline{6} \end{vmatrix}$$
$$= \underline{i} \begin{vmatrix} \underline{3} & \underline{-2} \\ \underline{s} & \underline{6} \end{vmatrix} - \underline{j} \begin{vmatrix} \underline{s} & \underline{-2} \\ \underline{-2} & \underline{6} \end{vmatrix} + \underline{k} \begin{vmatrix} \underline{s} & \underline{3} \\ \underline{-2} & \underline{s} \end{vmatrix}$$
$$= 28\underline{i} - 26\underline{j} + 31\underline{k}$$

b) using $\sin \alpha = \frac{\underline{d} \cdot \underline{n}}{|\underline{d}| |\underline{n}|}$, $\underline{d} = \begin{pmatrix} \underline{3} \\ \underline{1} \\ \underline{-1} \end{pmatrix}$, $\underline{n} = \begin{pmatrix} \underline{28} \\ \underline{-26} \\ \underline{31} \end{pmatrix}$

$$\Rightarrow \sin \alpha = \frac{84 - 26 - 31}{\sqrt{3^2 + 1^2 + 1^2} \sqrt{28^2 + 26^2 + 31^2}} = \frac{27}{\sqrt{26631}}$$

$$\alpha = 9.52^\circ$$

Could use $\cos \beta = \frac{27}{\sqrt{26631}}$ for $\beta = 80.48^\circ$
so $\alpha = 9.52^\circ$

Q7. $\underline{r} = \begin{pmatrix} \underline{4} \\ \underline{0} \\ \underline{0} \end{pmatrix} + t \begin{pmatrix} \underline{3} \\ \underline{-1} \\ \underline{2} \end{pmatrix}$, $6x - y + 3z = 80$

$$\underline{d} = \begin{pmatrix} \underline{3} \\ \underline{-1} \\ \underline{2} \end{pmatrix} \quad \underline{n} = \begin{pmatrix} \underline{6} \\ \underline{-1} \\ \underline{3} \end{pmatrix} \quad \checkmark \text{ coefficients}$$

$$\text{using } \sin \alpha = \frac{\underline{d} \cdot \underline{n}}{|\underline{d}| |\underline{n}|} = \frac{18 + 1 + 6}{\sqrt{3^2 + 1^2 + 2^2} \sqrt{6^2 + 1^2 + 3^2}} = \frac{25}{2\sqrt{161}}$$

$$\text{so } \alpha = 80.1^\circ$$

Advanced skills



EXAM PAPERS PRACTICE

$$Q8. \quad \underline{r} = \begin{pmatrix} 2+t \\ 3-2t \\ 4t \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$$

direction of line d

$$\underline{r} = \underline{a} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

IN-Plane vectors

$$\text{so } \underline{n} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 0 & 1 \\ 1 & -2 & 1 \end{vmatrix}$$

$$= \underline{i} \begin{vmatrix} 0 & 1 \\ -2 & 1 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & 0 \\ 1 & -2 \end{vmatrix}$$

$$= 2\underline{i} - 0\underline{j} - 2\underline{k} = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix}$$

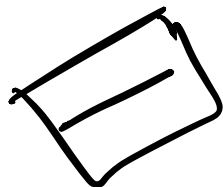
$$\text{now } \sin \alpha = \frac{\underline{d} \cdot \underline{n}}{|\underline{d}| |\underline{n}|} = \frac{2-8}{\sqrt{1^2+2^2+4^2} \sqrt{2^2+2^2}} = \frac{-6}{2\sqrt{42}}$$

$$\alpha = -27.6^\circ$$

$$\text{so } \alpha = 27.6^\circ$$

negative is just other direction

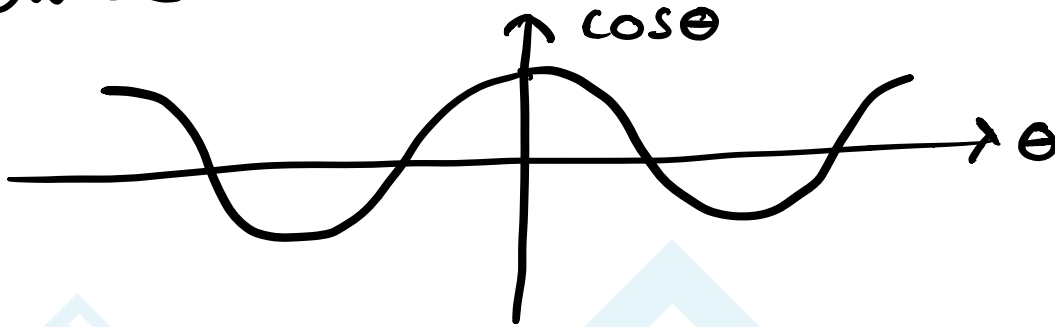
Q9.



for a line in plane the angle is zero hence $\underline{d} \cdot \underline{n} = 0$

Q10.

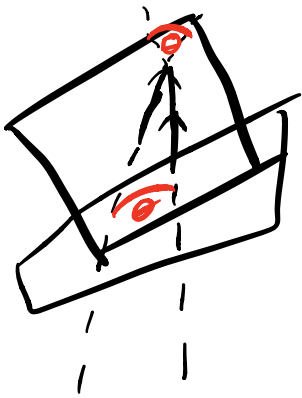
Amendment to limit $-1 \leq \cos \theta \leq 1$,
Otherwise it is correct



Q11. $5x - y = 11$ has $\underline{n}_1 = \begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix}$

$6x - 8y + z = 0$ has $\underline{n}_2 = \begin{pmatrix} 6 \\ -8 \\ 1 \end{pmatrix}$

Angle is same as between
normals



$$\text{SO } \cos \theta = \frac{\underline{n}_1 \cdot \underline{n}_2}{|\underline{n}_1| |\underline{n}_2|} = \frac{30 + 8 + 0}{\sqrt{3^2 + 1^2} \sqrt{6^2 + 8^2 + 1^2}}$$

$$\cos \theta = \frac{38}{\sqrt{2626}} \quad \theta = 42.1^\circ$$